

Experiment Schedule

Which study is completed when, and what it is for

Rev. 7, progress update: Sunday 4 October, afternoon

Companion to SHARC_Experiment_Programme.tex (rev. 7), which holds the method, metrics and criteria of every study. Experiments stop Friday 9 October; thesis writing 10–27 October.

Progress at Sunday 4 October, afternoon. Sunday's exit condition is met: every MUST study is complete except the E0 register (due Tuesday), and the SHOULD study V1 is also done. M1 passed in all four stress cases; V1 separated forward-model error from estimator error: truncation [50 10 10] is statistically too coarse for real data ($d_{\text{sys}} = 0.77$ to 3.22), depth $\pm 5\%$ is benign, and the common radial sensor position must be known or jointly estimated to sub-millimetre precision ($d_{\text{sys}} = 0.5$ at 0.05 mm). **Still open:** E0 (moved to Tuesday, done in one sitting); H1 co-located T and gravity-on- x coherence. Sunday afternoon is rest; Monday is V2 then D3.

1 At a glance (in working order)

Compressed plan (rev. 2): calibration logging starts today, the engine is built and validated by Friday morning, and every MUST study is finished by **Sunday 4 October**. Monday to Friday absorb slip, the stretch item, the register, figures and the freeze.

#	ID	Study	Goal	Tier	Day	Status
1	PR	Pre-registration	Fix scenarios, admissibility rules and class thresholds before any result exists	MUST	Wed 30	✓ Done
2	H1	Calibration: logging	Static orientations and translation excitation; gravity-on- x hold overnight	SHOULD	Wed 30	✓ Done
3	E1	Analytic engine: build	Scenario to metrics without Monte Carlo	MUST	Thu 1	✓ Done
4	H1	Calibration: analysis	Can real sensors meet 0.42% gain and 0.37 ms timing?	SHOULD	Thu 1	○ Partial
5	E0	Inventory: start	Register rows C1–C13 (moved: started and completed together on Tuesday)	MUST	Tue 6	Pending
6	E1	Analytic engine: validation gate	Reproduce U1/G3; earns the right to use the analytic route	MUST	Thu 1	✓ Done
7	R4	γ uncertainty	How accurately draught/thickness must be measured	SHOULD	Thu 1	✓ Done
8	D2	Record duration	How long to record for 1% β ; which L is feasible	SHOULD	Fri 2	✓ Done
9	R3	Noise-PSD mis-specification	How accurately the noise floor must be known	SHOULD	Fri 2	✓ Done
10	D1	Sensor count and layout	What each extra sensor buys; best layout and reference; Gramian vs Fisher	MUST	Fri 2	✓ Done
11	A1	Floe-property sweep	Which floe classes are identifiable	MUST	Sat 3	✓ Done
12	R2	Spectral shape	Sea-shape sensitivity of $\hat{T}$; is $L = 2048$ still adequate	MUST	Sat 3	✓ Done
13	R1	SNR and measured noise	Where real sensors sit on the noise axis; class boundaries	MUST	Sat 3	✓ Done
14	M1	Monte Carlo spot checks	Confirm analytic predictions away from p_0	MUST	Sun 4	✓ Done
15	V1	Forward-model mismatch	Separate estimator error from model discrepancy; truncation ladder first (A1 finding)	SHOULD	Sun 4	✓ Done
16	V2	Directional forcing	Bias from a second wave direction; can coherence detect it (promoted: D3 needs its forcing model)	SHOULD	Mon 5	Pending
17	D3	Direction-robust sensor count and layout	D1 repeated with unknown heading and V2's multi-directional forcing; does $n_s = 2$ still suffice	SHOULD	Mon 5	Pending
18	E0	Inventory: complete	Every completed study citable, figures flagged	MUST	Tue 6	Pending
19	–	Thesis-grade figures	Regenerate flagged figures; stretch items if time	SHOULD	Wed 7	Pending
20	FZ	Freeze	Every figure and number traceable; results summary	MUST	Fri 9	Pending

Row colour: done partly done white: pending. Day shows when an item was actually completed (the gate and R4 were planned for Friday 2). Row 17 (D3) is new in rev. 4. M1 ran overnight Saturday and was written up Sunday morning; V1 (stage 0 ladder, then the study) ran and was written up Sunday. H1 analysis: the 10-fold diagonal fit is done (z gain errors 0.10 to 0.44%, scale uncertainty 0.16 to 0.26%); the co-located transmissibility and gravity-on- x coherence are not yet.

Dependency rule. Only E0 and H1 are independent of E1. Everything from row 7 onward needs the E1 gate. M1 needs D1, A1 and R2 to choose its scenarios, so it launches Saturday night. D3 has two parts: D3a (unknown single heading) needs only D1's tools and can be pulled forward into any gap; D3b (multi-directional sea) needs V2's forcing model.

2 Day by day

Wednesday 30 September ✓ **complete**

- **PR, pre-registration** **MUST** A1 scenario list, R2 sea states, D1 admissibility, class thresholds, written into `defineStudyScenarios.m` as data only.

- **H1 logging** **SHOULD** (about 1.5 h, can run while PR is written):
 - one log per orientation, named `imu_cal_NN_<face>.log`, 12–18 orientations including the six faces, each held 3 min (the Allan minimum is at 100–400 s); tape the USB cable so it cannot pull the board;
 - one 10 min log of *vertical translation* excitation (board flat on a plate, moved up and down, not rocked), energy around 0.5–1.3 Hz;
 - **overnight**: board on its side with gravity along x (H1c).
- **Shell-rate dependence check** (15 min), suggestive only.
- **Message Justin**: PCB sensor positions and axis orientations, configured ODR and filters, meaning of the logged read-order offset.
- Send the repository (without `Data/Raw`) so the engine scaffolding can be drafted.

Done when: `defineStudyScenarios.m` committed; calibration logs saved; gravity-on- x hold running; Justin messaged. ✓

Thursday 1 October (heavy) ✓ complete, plus Friday's gate and R4

- **E1 build** **MUST** field sampler at fixed physical sensor positions, Jacobian in $\ln \beta$, $\ln R$, $\ln \gamma$, full analytic Σ_k , bias projector, `evaluateScenario.m`.
- **H1 analysis** **SHOULD** while EMM solves run: ellipsoid fit after its synthetic test; co-located transmissibility before and after calibration; gravity-on- x coherence. The lever-arm correction waits for Justin's PCB positions if needed.
- **E0 start** in gaps.

Done when: Σ_k and F at p_0 ; H1 table (provisional if PCB positions are outstanding). ✓

Achieved Thursday: ✓ E1 engine built (nine functions) and frozen; ✓ E1 validation gate passed 16/16 in MATLAB (commit 1147d1f; log `Results/E1/E1_gate_20261001.txt`); ✓ 10-fold calibration fit (commit 4cad5c1); ✓ reporting suite: standard figure style, results folder layout, CSV, file manifest; `testReporting` 6/6; ✓ R4 run and written up: γ to $\pm 1.6\%$ for $d_{\text{sys}} \leq 0.5$, $\pm 10.6\%$ for 1% in β (LSM6DSV16X); determine γ by weighing; ✓ E1 and R4 summaries (`SHARC_E1_Validation.tex`, `SHARC_R4_Gamma_Tolerance.tex`). **Not done:** E0 inventory start; H1 co-located T and gravity-on- x coherence.

Friday 2 October (heaviest) ✓ complete except E0

- ✓ **E1 validation gate** **MUST** done Thursday (16/16).
- ✓ **R4** **SHOULD** done Thursday.
- ✓ **D2, then R3** **SHOULD** quick wins, one at a time (run in MATLAB, then interpret and write up).
- **E0 start** (carried over from Thursday).
- ✓ **D1, sensor count and layout** **MUST** exhaustive $n_s = 2, 3, 4$, reference optimised, A/D/E criteria, Gramian comparison.
- ✓ **Overnight**: A1 EMM batch with truncation and node-grid checks (runner to be written first).

Done when: D2, R3 tables and summaries; D1 figures; A1 batch running. (Gate and R4 already done.)

Achieved Friday: ✓ D2 (commit dc8ee50): $\sigma_{\ln\beta} \leq 1\%$ needs 17.1 min at $L = 2048$ (LSM6DSV16X), feasible window 17.1 to 281.5 min; $L = 512$ is never Identifiable; the datasheet-grade case is Marginal at 109 min because d_W grows as $\sqrt{N}$; ✓ R3: the noise floor must be known to $\pm 30.6\%$ ($d_{\text{sys}} \leq 0.5$); the correction is required (none gives $d_{\text{sys}} = 1.63$); the measured noise colour is negligible; ✓ D1 (run D1_20261002_145943, gate exact to 10^{-12} , tests 10/10): two sensors suffice (s6 + reference s18, $\sigma_{\ln\beta} = 0.46\%$, Identifiable); gains -19.6% and -10.2% for the third and fourth sensors, below the 20% rule; optima nested on the incidence axis; reference choice irrelevant (median $\times 1.00$); Level 2C rank 167 of 40 920 ($\eta = 0.89$); Gramian designs $\eta = 0.21$ to 0.35 ; extra sensors buy placement robustness (layouts within 10% of optimum: 4, 13, 101). Four standard figures and printed symmetry and rotation checks; ✓ summaries SHARC_D2_Record_Duration.tex, SHARC_R3_Noise_Floor.tex, SHARC_D1_Sensor_Layout.tex; ✓ D3 defined (programme rev. 4). ✓ A1 batch runner written (resumable, checkpointed, with the model checks) and the batch run overnight. **Not done:** E0 inventory start.

Saturday 3 October (heavy) ✓ complete except E0

- ✓ A1, floe-property sweep **MUST** evaluate the batch; identifiability map.
- ✓ R2, spectral shape **MUST** and ✓ R1, SNR and measured noise **MUST**.
- ✓ Choose the four M1 scenarios; **write the M1 runner and launch it overnight** (100 records each).

Done when: A1 map; R1/R2 figures; M1 running.

Achieved Saturday: ✓ A1 (run A1_20261003_091530, tests 12/12): 12 Identifiable, 6 Marginal, 6 Not identifiable (class robust to the truncation). Lower boundary physical, between $R/\ell_f = 0.96$ and 1.64 (stiffness lost, radius kept); upper boundary estimator-limited at $kR \approx 3.7$ ($\sigma_{\ln\beta} \approx 0.2\%$ but $d_W \approx 0.6$); kR sets the precision in between. The SHARC model floe cannot yield a stiffness; a basin stiffness test needs a thin plate of about 1 m diameter. Truncation: metrics converged, model not ($d_{\text{trunc}} \geq 0.5$ in 16 of 24); ✓ EMM damped-root fallback in dispersionRoots.m (found by ring_beta_x0.25; tests 3/3, original roots unchanged; commit a7a31df); ✓ R2 (run R2_20261003_130557, tests 10/10): all 16 seas Identifiable at $L = 2048$ ($\sigma_{\ln\beta}$ 0.161 to 0.766% at equal in-band variance); $L = 1024$ only 3 of 16; $L = 4096$ lowest RMSE for all 16; exact T and J identical for every sea; ✓ R1 (run R1_20261003_135701): Identifiable window 110.4 to 653.8 $\mu\text{g}/\sqrt{\text{Hz}}$, bias-limited below, precision-limited above; LSM6DSV16X, ICM-42688-P, fused and G3 inside, datasheet grade and LSM6DS3TR-C outside; 0% invalid bins; ✓ M1 scenarios fixed by their pre-registered rules: D1 $n_s = 2$ (s6, ref. s18); miz_R25_h1; $\omega_p = 3.5$, $\gamma_J = 7$; p_0 at LSM6DSV16X; ✓ summaries SHARC_A1_Floe_Domain.tex, SHARC_R2_Spectral_Shape.tex, SHARC_R1_SNR.tex; summary PDFs backed up in the repository (1_MY_SUMMARIES/). ✓ M1 code (verbatim G3 estimator core, scenario resolver, checkpointed runner; tests 12/12; commit 96e9b0a) and the run launched at 22:11 (76 min). The runner now refuses to start with uncommitted code. **Not done:** E0 inventory start. **Provenance note:** the R1 run stamped commit a18e358, but its code was committed afterwards; from M1 on the runner enforces a clean tree.

Sunday 4 October (medium) ✓ complete except E0

- ✓ M1 **MUST** results (borderline cases extended to 200 records automatically), plotter and summary.
- ✓ V1, forward-model mismatch **SHOULD**. **First** (from A1): extend the truncation ladder in M (vertical modes, which dominate) until one step changes the features by $d < 0.1$, and set `cfg.V1.truncationTruth` to that level as a logged change *before* V1 runs. [70 15 15] is not a valid truth ($d = 0.25$ to [90 20 20] at p_0). Then the depth and sensor-position errors as planned.
- **E0 start** (carried over twice; time-box it to the M1 wait).

Done when: all **MUST** studies **complete** except the E0 register (due Tuesday).

✓

Achieved Sunday: ✓ M1 (run M1_20261003_221127, commit 96e9b0a): all four cases pass every pre-registered criterion. Scatter emp/pred 0.94 to 1.09 (band 0.85 to 1.20); 95% coverage 0.925 to 0.965 (band 0.88 to 0.99); mean offsets consistent with the Welch-expected bias (all within 1.4 SE); all eight nonlinear inversions converged from $\beta \times 0.8$ and $\times 1.2$ and agree with the linearised estimate within 0.018σ . Weakest case: the Froude-scaled MIZ floe, scatter 8 to 9% above prediction (about 1.5 SE), its A1 class unchanged. A1 and R2 were extended to 200 records by the borderline rule; ✓ plotM1MonteCarloSpotChecks.m (three figures) and SHARC_M1_Spot_Checks.tex; ✓ V1 stage 0 (truncationLadder_20261004_113129): $d = 0.125$ for M 90 to 110, 0.079 for 110 to 130, so the truth is [130 20 20]; converged also at the three large- kR floes ($d \leq 0.069$) and in P ($d = 0.050$); ✓ V1 (run V1_20261004_121316, commit 1711fb3, 59 min, tests 11/11): truncation [50 10 10] against [130 20 20] gives $d_{\text{sys}} = 0.77$ to 3.22 for sub-percent parameter shifts, moving p_0 to Marginal and two large- kR floes to Not identifiable, so real data need [130 20 20]; depth $\pm 5\%$ benign ($d_{\text{sys}} \leq 0.205$, tolerance $> \pm 5\%$); common radial sensor error $d_{\text{sys}} = 0.5$ at 0.050 mm, 1 at 0.10 mm, 1% in β at 0.29 mm, so sensor coordinates relative to the model geometry must be known or jointly estimated to sub-millimetre precision; the nonlinear inversion at -20 mm ($+80.5\%$ against linear $+107\%$) confirms centimetre errors are far outside the linear regime; ✓ plotV1ModelMismatch.m (three figures) and SHARC_V1_Model_Mismatch.tex. **Not done:** E0 inventory start (fourth carry-over).

Sunday afternoon: rest. V2 moves to Monday with D3; E0 moves to Tuesday.

Monday 5 October (heavy)

- **V2, directional forcing SHOULD** (promoted from STRETCH), **morning**. Directional forcing model first, as a reusable function (directionalSpectralMatrix.m: spectral matrix for components at heading ϕ and energy fraction, and for a spread) with its tester, because D3 calls it; then the bias sweep, then coherence detectability.
- **D3, direction-robust sensor count and layout SHOULD**, **afternoon**. D3a first (heading unknown: nuisance and worst case; needs only D1's tools), then D3b (exhaustive $n_s = 2, 3, 4$ under the V2 sea states with the directional bias in d_{tot}). Efficiency of the D1 optimum and of Level 2C under each; revisit n_s with the 20% rule. D3b can run overnight if the afternoon runs out.

Done when: V2 bias and detectability tables; forcing model callable; D3 recommendation (D3b may finish overnight).

Tuesday 6 October (medium)

- **E0 register MUST**, start and finish (moved from Sunday; one sitting, hard time-box half a day): rows C1–C13 plus one row per completed study (R4, D2, R3, D1, A1, R2, R1, M1, V1, V2, D3), figures flagged for regeneration.
- D3 write-up if D3b ran overnight; V2 and D3 summaries.
- Buffer: H1 with Justin's PCB positions (V1 makes the on-board sensor positions matter more, not less); *optional* V1 diagnostic (about 20 min, logged as supplementary): one nonlinear solve at $+0.1$ mm.

Done when: register complete; D3 written up; all experiments except figures and freeze done.

Wednesday 7 and Thursday 8 October (light)

- **Wed:** regenerate flagged figures to thesis grade.
- **Thu:** buffer; stretch items ($n_s = 5$, on-board fusion) only if time remains. A natural place to start the thesis results chapter outline.
- **Optional logged diagnostic** (in any gap, minutes): the large- kR A1 floes and the datasheet-grade R1 case at $L = 4096$, to test the hypothesis that a finer Welch resolution removes the estimator-limited boundary. Labelled as a diagnostic, not a change to A1 or R1.

Friday 9 October

- **FZ, freeze MUST**: every figure and number traceable; one-page results summary.

Done when: experiments stop.

3 Cut order if time runs short

1. Stretch items ($n_s = 5$, fusion), the $L = 4096$ diagnostic and the optional V1 +0.1 mm solve. (Extended M1 is no longer needed: all four cases passed.)
2. V2 coherence-detectability part (keep the forcing model and the bias sweep, which D3 needs).
3. D3b reduced to $n_s = 2, 3$ and two V2 sea states; keep D3a (unknown heading), which needs no V2 work. (First cut if Monday overruns, since V2 and D3 now share one day.)
4. H1 ellipsoid fit (keep co-located T before calibration and the gravity-on- x hold).
5. A1 corners and R3 (*already done*).

E1, D1, A1, R1, R2, M1, V1 (all done) and the freeze are never cut. D3a is cheap (permutations of the D1 field for 45° steps) and should survive any cut that leaves a free afternoon.